



1. Introduction (For Easy Reference)

The **BT401 module** is a 5-in-1 solution that supports Bluetooth, U-disk, and TF card playback. The module's highlights include support for dual-mode Bluetooth, meaning Bluetooth audio and Bluetooth data can operate simultaneously, along with simple and clear **serial port control functions, supporting BLE pass-through and SPP pass-through capabilities**. This greatly reduces the development difficulty of embedding Bluetooth in other products. **Note: The module uses a stamp-hole design with holes on both sides; for details, see "Chapter 11". It also has a very convenient firmware upgrade interface; refer to "Chapter 9".**

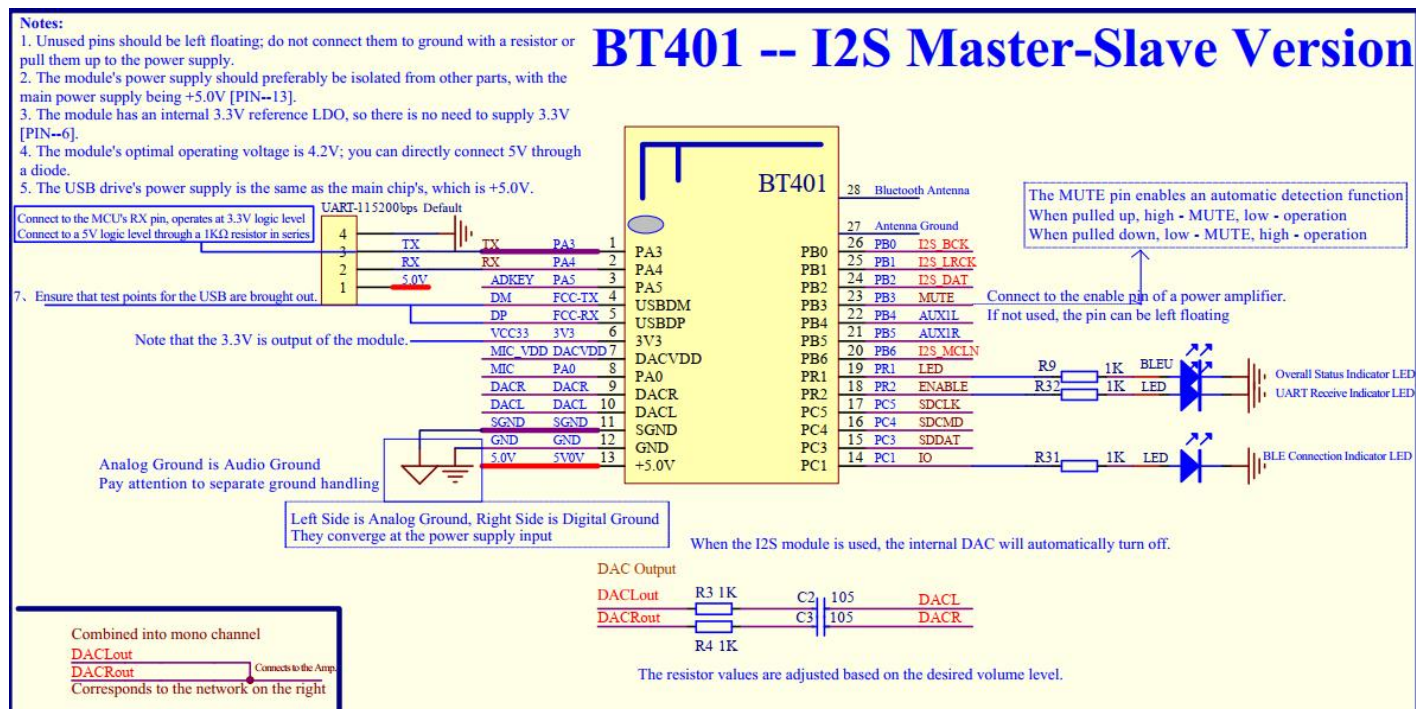
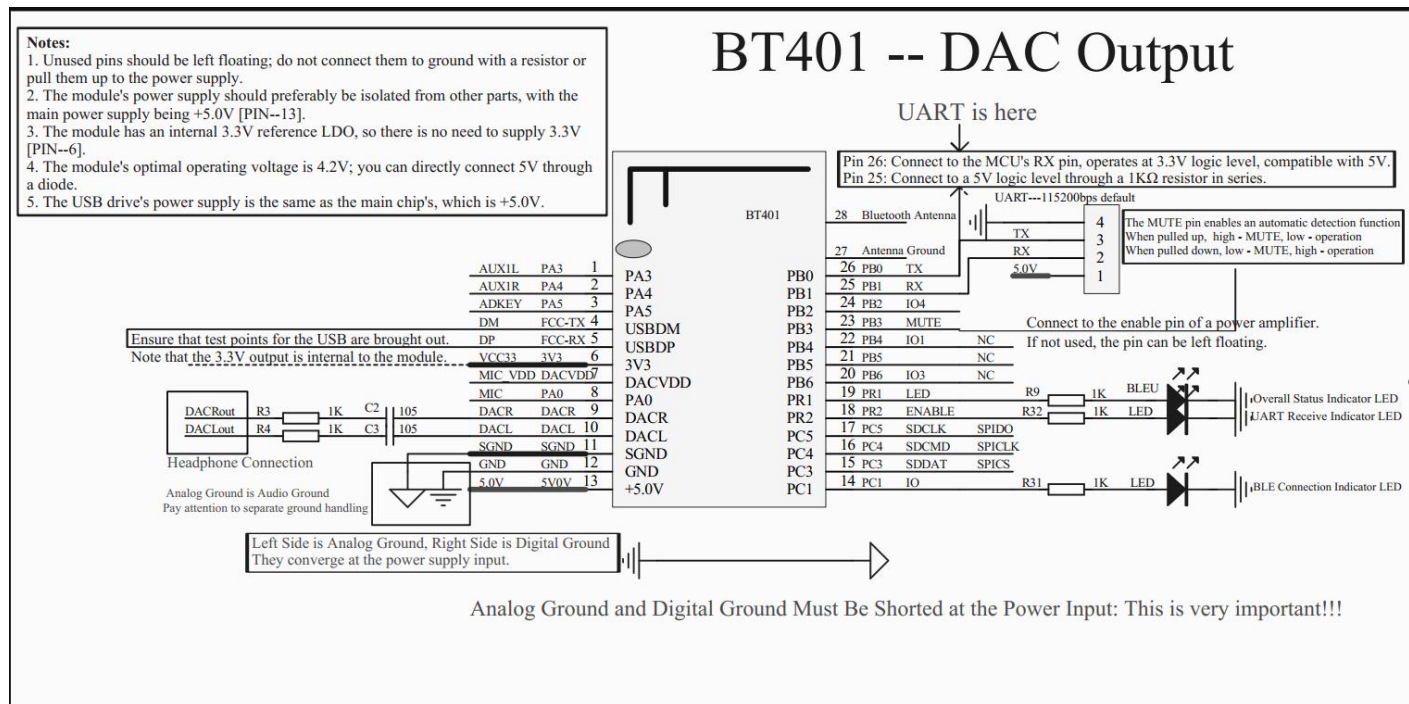
Function Introduction

1. Supports Bluetooth audio, **BLE data transmission**, U disk playback, and TF card playback. Supports decoding of MP3, WAV, FLAC, and APE formats.
2. **Optional**: Serial port AT command control. Supports modifying baud rate, changing Bluetooth name, specifying device playback, and data transparent transmission with a mobile phone.
3. Supports dual-mode data transmission with BLE and SPP, while also allowing Bluetooth audio playback without conflicts.
4. Supports Bluetooth calls, multi-button control, and simple functions such as previous track, next track, play/pause, and volume adjustment.
5. Supports retrieving the complete file name of the played file, specifying playback by serial port path, and looping or single playback.
6. Supports 128Kbps recording function. Records to TF card or U disk, with features like recording deletion and playback.
7. Comprehensive usage materials, including reference PCB, usage video tutorials, mobile phone transparent transmission video tutorials, and serial port AT command video tutorials.
8. Ultra-low cost, competitive pricing for large quantities, highly flexible firmware upgrades, and support for DAC or I2S audio output with a 48kHz sampling rate.
9. Easy integration with existing apps or development of WeChat mini-programs to enable mobile phone control.

1. Bluetooth Audio Products	E.g., Bluetooth speakers, Bluetooth headsets, car Bluetooth systems, etc.
2. Bluetooth Data Transmission Products	E.g., smart door locks, car OBD diagnostics, smart carts, printers, medical device data collection.
3. Bluetooth Data Transmission + Audio Products	E.g., Bluetooth music lights, Bluetooth broadcasting, electronic keyboards, and other musical instruments.
Note: This solution is not suitable for applications requiring ultra-low power consumption.	



2. Schematic diagram (For Easy Reference)



Note: There are two basic versions here: DAC output and I2S digital audio version.

DAC Version: The BT401 Bluetooth module [pins 9 and 10] directly outputs analog audio signals. In other words, it can directly drive headphones to produce sound.

I2S Version: The BT401 Bluetooth module [pins 24, 25, 26] directly outputs or inputs digital signals, typically used with a DSP.

If you don't understand what I2S is, just focus on the DAC version. If you want to learn about I2S, search online yourself; no training is provided.

3. Minimum System (For Easy Reference)

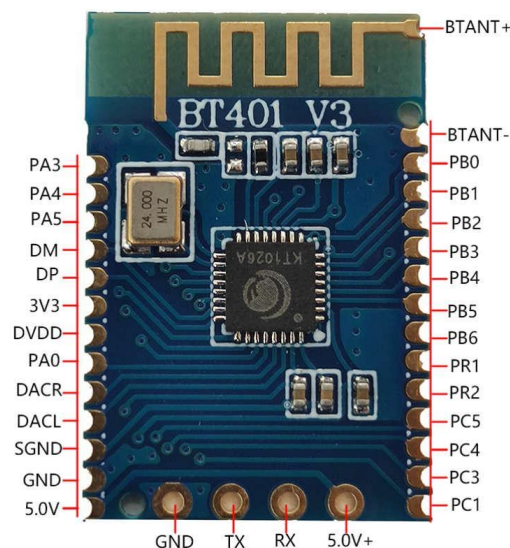
Test Board



Front



Back



When getting samples, we recommend purchasing a test board blank PCB at the same time - this way you'll have a reference for testing.

Although it's simple, and in fact using it is simple too, you can never rule out various unexpected situations that might waste your debugging time. Always use the test board as your reference standard.

 BT401 Bluetooth Module Complete Data Download Link: -- May 23, 2024
 Netdisk: https://drive.google.com/drive/folders/1QD7j4i9X_mTUXQqpZ-9S1MSmpFrfs5J8?usp=drive_link
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4. FAQ Collection (Remember Keyword Search)

Question 1	1. What is BLE 4.0?
Answer	Starting from Bluetooth 4.0, there are two branches: Classic 4.0 and BLE 4.0. Classic 4.0 is an upgrade from the traditional Bluetooth 3.0 and is backward compatible. BLE 4.0, however, is a new branch that is not backward compatible. BLE stands for Bluetooth Low Energy, and as the name suggests, it has lower power consumption. Our chip solution is Bluetooth 5.0, which is currently the latest version.

Question 2	2. Which devices support BLE?
Answer	iOS Platform: Devices such as iPhone 4s, 5, 5c, 5s, iPad 3, 4, and mini support BLE, with no need for MFI certification. Android Platform: Android phones equipped with Bluetooth 4.0 or higher and upgraded to Android 4.3 or a higher system version.

Question 3	3. Why can't I find the BLE device in the system Bluetooth interface?
Answer	Phone Bluetooth operates in classic mode by default. You need to use an app to handle the entire process of searching, pairing, connecting, and communicating. For iOS, if you don't have a developer certificate, please download LightBlue from the Apple Store. For Android, you can use the simple BLE serial assistant provided by us for basic debugging, or test with your own app software.

Question 4	4. What is the transmission speed of BLE?
Answer	The physical bandwidth of BLE is only 1Mbps, with actual transmission speeds ranging from 1 to 6KB/s. The specific speed needs to be verified based on your usage environment and project. If possible, please use a higher baud rate whenever feasible, as it will provide better communication performance.

Question 5	5. Are there byte restrictions for sending and receiving?
Answer	The BLE module provided by us has no byte restrictions for receiving or sending via the serial port. On iOS systems, the module currently supports a maximum transmission of 90 bytes per packet. However, research shows that iOS internally splits packets with low efficiency. Therefore, please split data exceeding 20 bytes into smaller packets for transmission. Our module has been optimized in this regard, improving efficiency by 10% compared to iOS's internal packet splitting.



Question 6	6. How to troubleshoot common issues with the Bluetooth module?
Answer	1. Every module is programmed with debugging code via the serial port before leaving the factory, ensuring the serial port has no issues. If a problem occurs, don't panic or doubt the module's serial port functionality; the serial port is 100% qualified at the factory. The module's serial port uses TTL levels, so if connecting to a computer, use an RS232 level conversion chip or corresponding circuit. When sending AT commands, always include a carriage return and line feed at the end. Some serial assistants have a "Send New Line" option, which must be enabled. 2. Do not search for Bluetooth BLE in the phone's settings, as the phone system defaults to classic mode and does not support searching for BLE 4.0 devices.

Question 7	7. The documentation has so much information, and I can't understand it. I only need a module that can connect to a phone to play music, support calls, and have button control.
Answer	That's fine. Our default functions can support this. Standardizing the product allows us to meet most customers' needs and reduce costs. So, extra functions don't mean it's more expensive; in fact, it's cheaper. Other customers may use those functions, but you don't need them. Just ignore the parts you don't understand. Multiple functions won't cause any conflicts. You can skip the serial port section.

Question 8	8. Can this module connect to a phone to play music? Does it support calls?
Answer	Yes, the module supports Bluetooth audio and Bluetooth calls [calls are enabled by default, but users can send a command to disable them]. Additionally, it supports data transparent transmission.

Question 9	9. Your module is too large to embed in our product. Can I purchase the chip directly? Is the peripheral circuitry complex?
Answer	Yes, our module is designed to be convenient for most customers to use. The board is designed this way to make testing very convenient and efficient for customers. If you need the chip, you can contact us, or we can customize a suitable module based on your company's needs.

Question 10	10. I don't need the call function. How can I handle this?
Answer	You can disable the call function via a serial port command. After the next power-up, Bluetooth will no longer support calls. The same applies to re-enabling it. In the manual, search for the keyword "call" to find the relevant instructions.



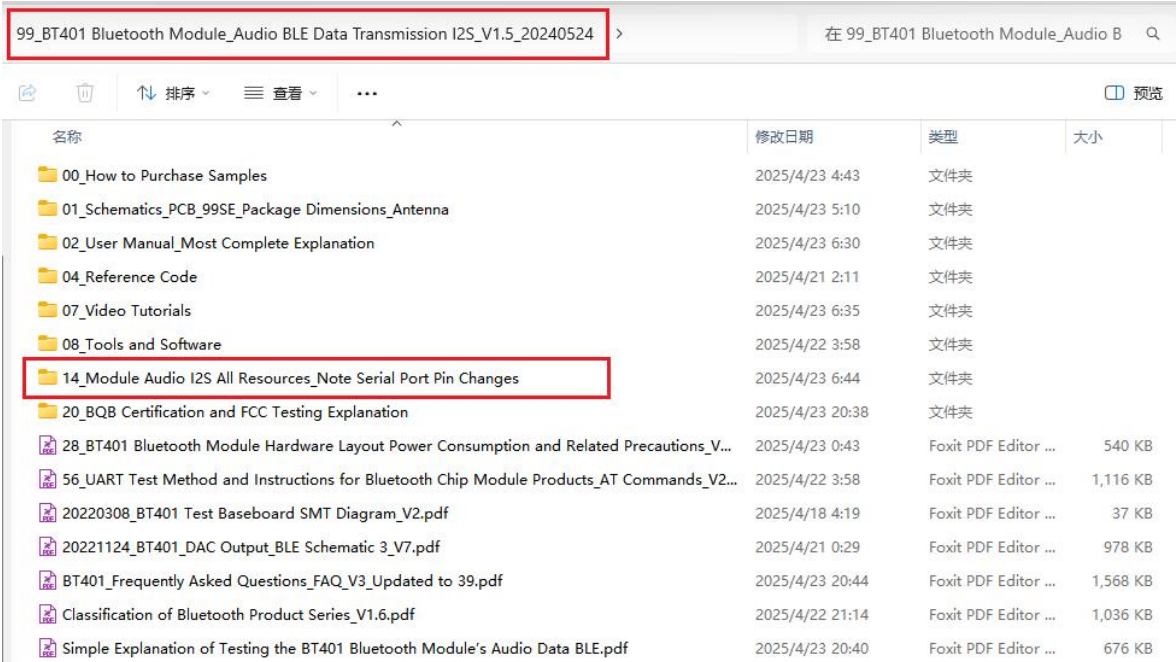
Question 11	11. I don't need the BLE data function. How can I disable it?
Answer	Yes, you can simply send a serial port command to disable BLE. After the next power-up, the chip will not initialize the BLE protocol stack. Search for the keyword "BLE" in the manual.

Question 12	12. I don't need the music function, only BLE or SPP transparent transmission. How can I do this?
Answer	You can disable Bluetooth audio via a serial port AT command. Check the manual for details. The same applies to re-enabling it. Search for the keyword "audio" in the manual.

Question 13	13. I only need Bluetooth music playback and calls, not TF card or U disk. What should I do?
Answer	Simply don't connect the related circuits on the hardware side. Our system has automatic detection, and if you don't use them, the software handles it internally. If you only want to use the TF card and not Bluetooth, that's also possible. Set it to automatically read the device on power-up, and if no device is present, it will switch to Bluetooth.

Question 14	14. What does Bluetooth transparent transmission mean in the documentation?
Answer	Bluetooth transparent transmission means: Whatever the MCU sends to Bluetooth, Bluetooth sends to the phone. Conversely, whatever the phone sends to Bluetooth, Bluetooth sends to your MCU via the serial port. The chip does no data processing; it simply acts as a relay. The BT401 supports dual-mode transparent transmission.

Question 15	15. Why does my phone only show "BT401-AUDIO" when searching for Bluetooth and not "BT401-BLE"?
Answer	The phone's system settings can only detect classic Bluetooth, which is "BT401-AUDIO." It cannot detect BLE devices. To test BLE, please use the LightBlue test app to search. In the "02_User Manual" document package, there is a detailed manual and many videos to help customers understand and test. Please take a look.

Question 16	16. Why is the serial port data garbled when I power on the I2S digital audio version of the module
Answer	For the I2S version of the module, the serial port is on pins 1 and 2. You can check the I2S schematic for details. See the detailed documentation in the document package, as shown in the figure below: 

Question 17	17.Why does my module keep making a “beep beep beep” sound after powering on?
Answer	Please check if pin 3, the ADkey pin, is floating. Normally, this pin should have a 3.3V voltage.

Question 18	18. Are the PCB layout requirements for the BT401 module high?
Answer	1. Bluetooth technology is now very mature. Solutions with overly high production requirements lack market competitiveness, so it's definitely simple—don't worry. 2. Just ensure the analog SGND is connected to the power input GND, and there won't be any issues. The GND of surrounding capacitors should be as close as possible to the chip's GND. 3. Place the Bluetooth antenna near the board's edge in an open area, avoiding metal enclosure. See the following document for details:



Question 21	21. What does the “checksum” marked on the Bluetooth module sample card mean?
Answer	1. The chip is programmable, so there are many firmware versions. Be sure to remember that as long as the checksum is correct, burning the firmware won't go wrong. 2. How do we distinguish firmwares? That's where the checksum comes in. 3. The checksum is unique, generated by our CRC checksum tool. 4. The checksum is critical information to remember, as we have many customers and software versions, and we rely on the checksum for accuracy. 5. During batch programming, we also use the checksum to ensure the burned firmware is correct.

Question 22	22. What are the ESD data and operating temperature of the BT401 module?
Answer	1. Contact discharge: 4kV; air discharge: 8kV. 2. Operating temperature range: -40°C to +85°C. The module's core parameters depend on the main control chip, and the above parameters are the chip's specifications.

Question 23	23. What is the amplitude and output level of the BT401 module's DAC output?
Answer	1. The output bias voltage is approximately 1.5V. 2. The fluctuation is 1.5V up and down, so $V_{pp} = 3V$. 3. The BT401's DAC output volume is quite high. You can get a test board, solder a headphone jack, plug in 3.5mm headphones, and listen—the sound is loud.

Question 24

24. Where are the materials or documents for the I2S version of the BT401 module?

Answer

99_BT401 Bluetooth Module_Audio BLE Data Transmission I2S_V1.5_20240524 > 在 99_BT401 Bluetooth Module_Audio B

排序 查看

名称	修改日期	类型	大小
00_How to Purchase Samples	2025/4/23 4:43	文件夹	
01_Schematics_PCB_99SE_Package Dimensions_Antenna	2025/4/23 5:10	文件夹	
02_User Manual_Most Complete Explanation	2025/4/23 6:30	文件夹	
04_Reference Code	2025/4/21 2:11	文件夹	
07_Video Tutorials	2025/4/23 6:35	文件夹	
08_Tools and Software	2025/4/22 3:58	文件夹	
14_Module Audio I2S All Resources_Note Serial Port Pin Changes	2025/4/23 6:44	文件夹	
20_BQB Certification and FCC Testing Explanation	2025/4/23 20:38	文件夹	
28_BT401 Bluetooth Module Hardware Layout Power Consumption and Related Precautions_V...	2025/4/23 0:43	Foxit PDF Editor ...	540 KB
56_UART Test Method and Instructions for Bluetooth Chip Module Products_AT Commands_V2...	2025/4/22 3:58	Foxit PDF Editor ...	1,116 KB
20220308_BT401 Test Baseboard SMT Diagram_V2.pdf	2025/4/18 4:19	Foxit PDF Editor ...	37 KB
20221124_BT401_DAC Output_BLE Schematic 3_V7.pdf	2025/4/21 0:29	Foxit PDF Editor ...	978 KB
BT401_Frequently Asked Questions_FAQ_V3_Updated to 39.pdf	2025/4/23 20:44	Foxit PDF Editor ...	1,568 KB
Classification of Bluetooth Product Series_V1.6.pdf	2025/4/22 21:14	Foxit PDF Editor ...	1,036 KB
Simple Explanation of Testing the BT401 Bluetooth Module's Audio Data BLE.pdf	2025/4/23 20:40	Foxit PDF Editor ...	676 KB

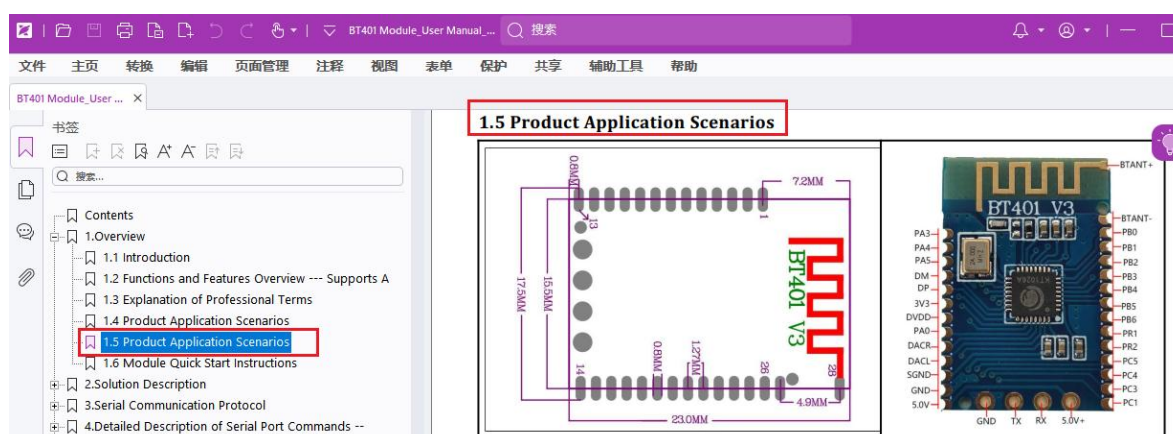
The digital audio materials or documents for the BT401 are in a separate document package, mainly covering hardware and usage instructions.
The AT commands are still in the manual in the 02 folder.

Question 25

25. Can you provide the BT401 module's package diagram? I need to modify the pads.

Answer

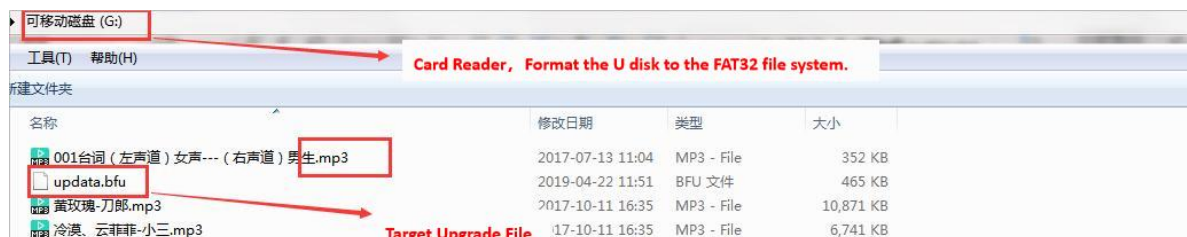
1. The BT401's package dimensions are detailed in two places: see Section 1.5 of the manual.



2. The second location is the reference PCB test baseboard, found in the following location: in the 01 folder of the document package.

99_BT401 Bluetooth Module_Audio BLE Data Transmission I2S_V1.5_20240524 > 01_Schematics_PCB_99SE_Package Dimensions_Antenna				
名称	修改日期	类型	大小	
00_Common Component Footprints_99SE	2025/4/23 4:57	文件夹		
01_BT401 Reference PCB and Schematics_99SE	2025/4/23 4:59	文件夹		
02_Amplifier Circuit_8002_HAA2018	2025/4/23 4:51	文件夹		
03_Module Encapsulation	2025/4/23 4:57	文件夹		
04_Bluetooth Antenna	2025/4/23 5:07	文件夹		
20191025_BT401_Dimension Diagram V3_External An...	2025/4/23 5:20	Foxit PDF Editor ...	2,558 KB	
BT401 Module Real Photos V3.jpg	2019/12/26 9:16	JPG 文件	100 KB	
readme.txt	2025/4/23 5:16	文本文档	4 KB	

Question 26	26. Explanation and Introduction to Firmware Upgrades for the BT401 Module, Including Support for Different Firmware Versions
Answer	1. Copy the “updata.bfu” file to a TF card or U disk. The TF card or U disk must contain at least 3 MP3 audio files for the file system to read. 2. Power on the prototype, then insert the U disk. The firmware will automatically update, and it will take approximately 30 seconds to complete. 3. Signs of a successful upgrade: If there is an indicator light, it will stay on during the upgrade and turn off when the upgrade is complete. 4. This method is now fully described. For machines with prompt sounds enabled, inserting a U disk or TF card will trigger a prompt sound. 5. Note: After the upgrade is complete, you must delete the upgrade file from the card, otherwise it will repeatedly attempt to upgrade. During the upgrade process, do not remove the TF card or U disk, as this will cause the module to completely crash and require factory repair.



Preferably use a U disk, especially one with an indicator light; the capacity is not important.

1. Copy the updata.bfu file to a TF card or U disk. The TF card or U disk must contain at least 3 MP3 audio files for the file system to read.
2. Power on the prototype, then insert the U disk. The program will automatically update, and it will take approximately 30 seconds to complete.
3. Signs of a successful upgrade: If there is an indicator light, it will stay on during the upgrade and turn off when the upgrade is complete.
4. This method is now fully described. For machines with prompt sounds enabled, inserting a U disk or TF card will trigger a prompt sound.
5. Note: After the upgrade is complete, you must delete the upgrade file from the card, otherwise it will repeatedly attempt to upgrade.

During the upgrade process, do not remove the TF card or U disk, as this will cause the module to completely crash and require factory repair.

On the computer side, it's best to enable the display of file extensions. Pay attention to comparing with this screenshot.

Upgrading is considered a value-added service and is not recommended. If the upgrade fails, the module will stop working. Don't come to us asking why it happened or how to fix it. Buy the version you need and avoid failures. It's a headache for us to analyze these issues.

1. Are there any requirements for the TF card and U disk?

The TF card and U disk must use the FAT or FAT32 file system, with a maximum capacity of 32GB.

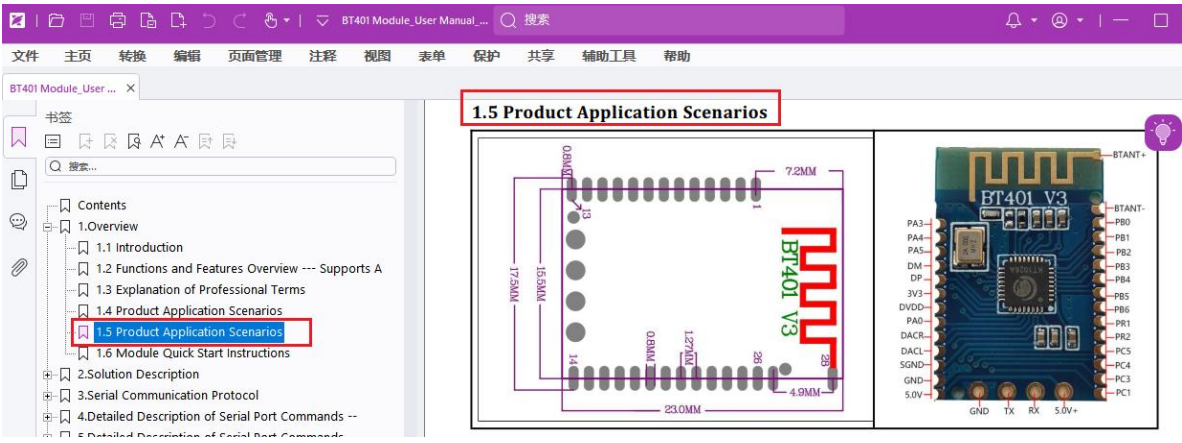
2. Why does my prototype start playing music directly when I copy the "updata.bfu" file to a TF card for upgrading and insert it into the prototype?

1. Please ensure the file name is exactly "updata.bfu". The name must match perfectly, with no extra or missing letters.
2. "Updata" is the file name, and "bfu" is the file extension.
3. When working on Windows, make sure to enable Show file extensions in the settings.

3. Why doesn't the prototype enter upgrade mode after I copy the "updata.bfu" file to a TF card, insert it, and power on the prototype?

Answer:

1. Our upgrade process requires you to power on the prototype first, then insert the TF card.
2. The upgrade principle is as follows (the same applies to a U disk):
 - (1) After the chip detects the TF card insertion, it automatically initializes the TF card and the file system.
 - (2) If initialization is successful, it searches for the "updata.bfu" file. If found, it enters upgrade mode.
 - (3) If any of these steps fail, the upgrade will not succeed.

Question 27	27. Does the BT401 module support external antennas, such as small chili antennas, 2.4G antennas, or IPEX connector antennas?
Answer	1. The BT401 Bluetooth module comes with an onboard antenna, and its performance is quite good. 2. However, for some enclosures, such as metal ones with higher requirements, an external antenna may be needed. The Bluetooth module also supports this. Details are shown in the figure below, where the last two pins, “BTANT+” and “BTANT-,” correspond to the positive and negative poles of the antenna. 
Question 28	28. What are the testing methods and instructions for the UART serial port commands of the Bluetooth chip module product, including changing the name, address, etc.?
Answer	We won't cover basic serial port concepts, as there's plenty of information online. The product's serial port uses AT commands, but note that they are non-standard and differ from other manufacturers' modules. The factory default baud rate is 115200. Upon power-up, the module automatically returns parameters such as version, Bluetooth name, UUID, etc. See “Section 3” of the manual for details; they are generally similar, with serial ports cross-connected. For a complete and detailed explanation, refer to another very detailed document in the document package. Please take a look.



99_BT401 Bluetooth Module_Audio BLE Data Transmission I2S_V1.5_20240524

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名称	修改日期	类型	大小
00_How to Purchase Samples	2025/4/23 4:43	文件夹	
01_Schematics_PCB_99SE_Package Dimensions_Antenna	2025/4/23 5:10	文件夹	
02_User Manual_Most Complete Explanation	2025/4/23 6:30	文件夹	
04_Reference Code	2025/4/21 2:11	文件夹	
07_Video Tutorials	2025/4/23 6:35	文件夹	
08_Tools and Software	2025/4/22 3:58	文件夹	
14_Module Audio I2S All Resources_Note Serial Port Pin Changes	2025/4/23 6:44	文件夹	
20_BQB Certification and FCC Testing Explanation	2025/4/23 20:38	文件夹	
28_BT401 Bluetooth Module Hardware Layout Power Consumption and Related Precautions_V...	2025/4/23 0:43	Foxit PDF Editor ...	540 KB
56_UART Test Method and Instructions for Bluetooth Chip Module Products_AT Commands_V2...	2025/4/22 3:58	Foxit PDF Editor ...	1,116 KB
20220308_BT401 Test Baseboard SMT Diagram_V2.pdf	2025/4/18 4:19	Foxit PDF Editor ...	37 KB
20221124_BT401_DAC Output_BLE Schematic_V7.pdf	2025/4/21 0:29	Foxit PDF Editor ...	978 KB
BT401_Frequently Asked Questions_FAQ_V3_Updated to 39.pdf	2025/4/23 20:44	Foxit PDF Editor ...	1,568 KB
Classification of Bluetooth Product Series_V1.6.pdf	2025/4/22 21:14	Foxit PDF Editor ...	1,036 KB
Simple Explanation of Testing the BT401 Bluetooth Module's Audio Data BLE.pdf	2025/4/23 20:40	Foxit PDF Editor ...	676 KB

Question 29

29. Can the default firmware of the purchased BT401 Bluetooth module directly connect to a phone to play music?

Answer

1. Yes, the default BT401 Bluetooth module DAC version can be powered on and found in the phone's Settings > Bluetooth menu. The default Bluetooth name is typically "BT401-audio."

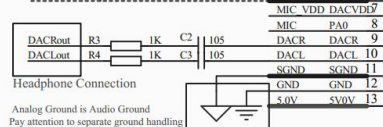
2. Also, note that pin 19 (PR1) of the Bluetooth module can be connected to an external indicator light to easily observe the status:

==> Searching = fast flashing; Connected = steady on; Playing = slow flashing.

Notes:

- Unused pins should be left floating; do not connect them to ground with a resistor or pull them up to the power supply.
- The module's power supply should preferably be isolated from other parts, with the main power supply being +5.0V [PIN--13].
- The module has an internal 3.3V reference LDO, so there is no need to supply 3.3V [PIN--6].
- The module's optimal operating voltage is 4.2V; you can directly connect 5V through a diode.
- The USB drive's power supply is the same as the main chip's, which is +5.0V.

Ensure that test points for the USB are brought out.
Note that the 3.3V output is internal to the module.



Headphone Connection

Analog Ground is Audio Ground

Pay attention to separate ground handling

Left Side is Analog Ground, Right Side is Digital Ground
They converge at the power supply input.

BT401 -- DAC Output

UART is here

Pin 26: Connect to the MCU's RX pin, operates at 3.3V logic level, compatible with 5V.
Pin 25: Connect to a 5V logic level through a 1KΩ resistor in series.

UART--115200bps default

The MUTE pin enables an automatic detection function
When pulled up, high • MUTE, low • operation
When pulled down, low • MUTE, high • operation

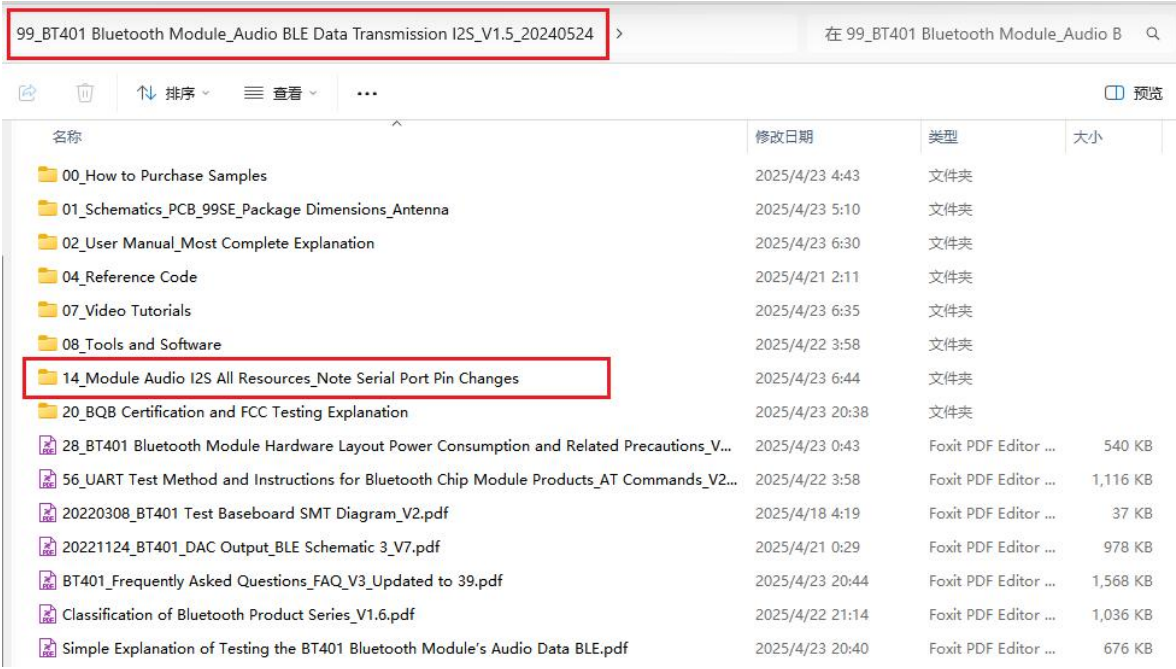
Connect to the enable pin of a power amplifier.
If not used, the pin can be left floating.

Overall Status Indicator LED

UART Receive Indicator LED

BLE Connection Indicator LED

Analog Ground and Digital Ground Must Be Shorted at the Power Input: This is very important!!!

Question 30	30. Are the pin assignments the same for the DAC version and I2S version of the purchased BT401 Bluetooth module?
Answer	1. Yes, their physical pin assignments are the same. However, their usage differs. 2. Most importantly, the serial port pin definitions are different. See Folder 14 in the document package for details.  3. For the DAC output version, the bottom four pins of the module are for the serial port. However, for the I2S version, they are not.

Question 31	31. The purchased BT401 Bluetooth module's functions are suitable, but how can I add customized functions based on this module?
Answer	1. We currently offer various chip models, including the LQFP48 package with multiple IO pins and the compact QFN32 package. These use the same die, so there are many possibilities depending on the customer's needs. 2. If customization is needed, please follow this process: (1) Clearly document your functional requirements. If there's uncertainty requiring discussion, contact our engineer Tsui at +86 15902797635 (WhatsApp) to confirm feasibility and keep communication efficient. (2) After clarifying details, provide us with a detailed document—the more detailed, the better. (3) Upon receiving your requirements, we will carefully evaluate them, provide a completion timeline, and discuss development details. Even if we can't meet your needs, we'll explain why and offer reasonable suggestions. (4) Custom solutions require a fee, which is unavoidable. The fee depends on the complexity of the functions, with no fixed standard. For a quote, please submit your

requirements.

If the cost poses a challenge, we're open to discussion and cooperation, depending on the sincerity of your product.

Question 32

32. What is the measured power consumption of the BT401 module's Bluetooth audio and BLE chip, and how much is the current?

Answer



With the status indicator light removed, tested with a 5V stabilized power supply—accuracy is good, with an error of about 1mA.

Bluetooth State:

1. Dual-mode audio + BLE -- Searching state: 32mA
2. Dual-mode audio + BLE -- Connected state -- Paused: 14mA
3. Dual-mode audio + BLE -- Connected state -- Playing -- No headphones (unloaded): 23mA
4. Pure BLE state -- Searching state: 9mA
5. Pure BLE state -- Connected state: 9mA

MP3 State:

1. Playing TF card -- Playing state: 43mA
2. Playing TF card -- Paused state: 38mA
3. Playing U disk -- Playing state: Depends on the U disk

Question 33

33. Power Supply Explanation for the BT401 Bluetooth BLE Audio Chip—Minimum, Maximum, and Most Suitable Supply Voltage

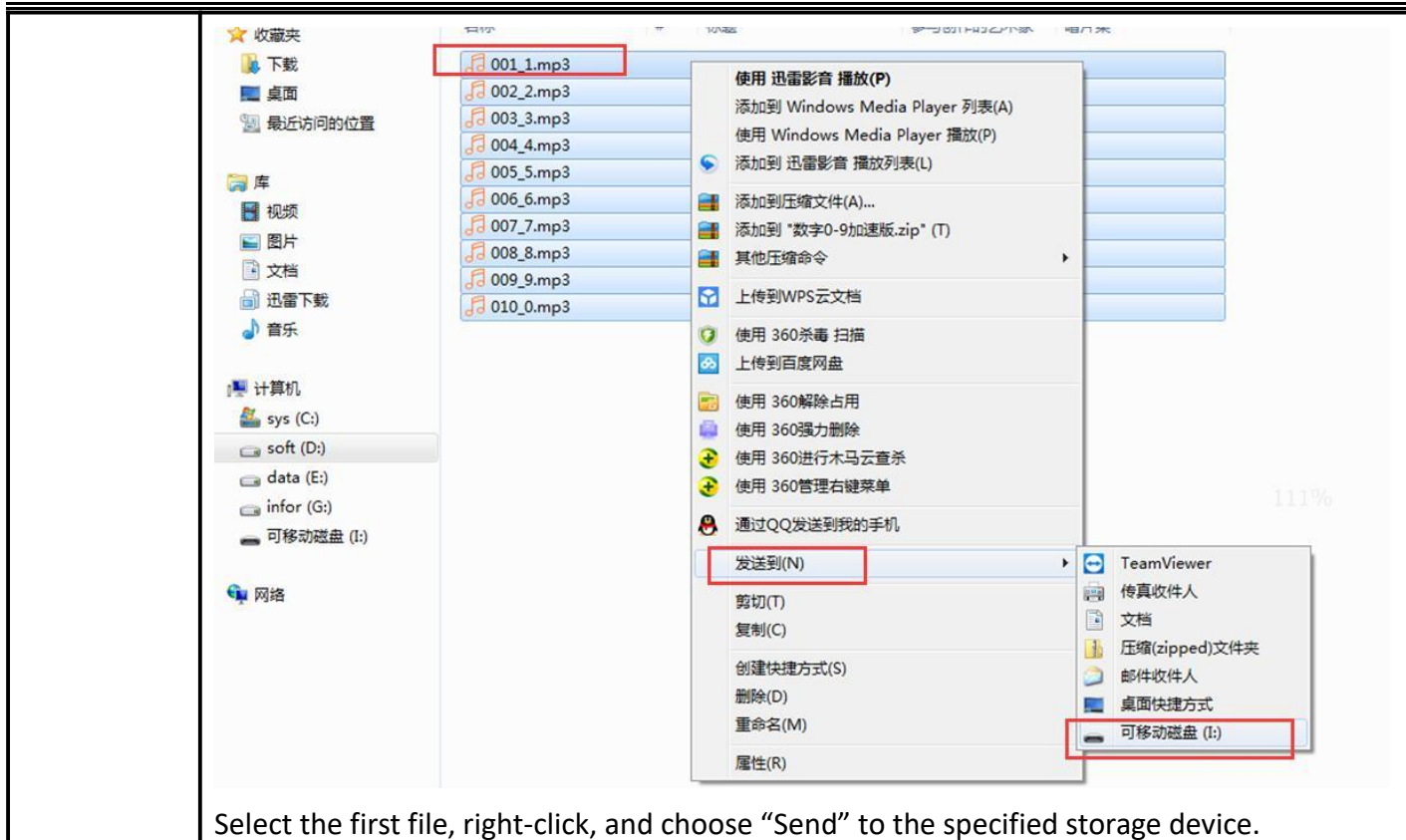
Answer

1. Pin 13 of the module is the chip's power supply input, supporting 3.2V to 5.2V.
2. Pin 6 is the module's internal LDO output—it's an output, pay attention!!
3. For pins you don't understand, just follow the schematics we provided and copy them.

About the chip's power supply and usage tips:

	1. The module is designed for “lithium battery” power scenarios, so pin 13 is named “VBAT,” meaning battery power. The chip's optimal working voltage is around 4.2V, but connecting directly to 5V is completely fine. 2. It supports a wide voltage range of 3.2V to 5.2V. Even slightly lower may work, but testing is required. 3. To support lithium battery power, the chip has a built-in 3.3V LDO to power the chip's core, so the IO operates at 3.3V levels. 4. Pin 12 is the LDO output. When pin 18's supply voltage exceeds 3.3V, pin 12 outputs 3.3V. If pin 18's input voltage is at or below 3.3V, pin 12 cannot output 3.3V and may be about 0.1V lower. 5. If you're not using an external power amplifier to drive a speaker, the chip works fine at 3.3V, even slightly lower. 6. However, if your system includes a speaker or horn, consider that sound output causes voltage fluctuations. 7. A 3.3V-powered system cannot drive large speakers, as a typical 4-ohm 3W speaker draws around 1A when producing sound, depending on the power amplifier chip. Please consider this carefully. Important Note: Even if the USB pins are not used, it's best to bring out test points. Please pay close attention to this.
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Question 34	34. Explanation of Copying to U Disk/TF Card/SPI Flash in Physical Order for BT401
Answer	Many customers need to copy files to the device in a specific order for playback. Our chip plays files in the physical order they were copied, meaning the first file copied is numbered 1, the second is numbered 2, and so on. This physical order is the “order in which you copied the files,” not the order shown in Windows. ==> It's best to rename the files so they are sorted in order on the computer. ==> Follow the operation shown in the figure below: Select the content to copy, place the mouse on the first file, right-click, and choose “Send.”



Question 35 35. Explanation of the BT401's MUTE Pin and the Purpose of Pull-Up/Pull-Down, with Detailed Description

Answer

id with a resistor or
her parts, with the
red to supply 3.3V
connect 5V through
ich is +5.0V.

AUX1L	PA3	1	PA3
AUX1R	PA4	2	PA4
ADKEY	PA5	3	PA5
DM	FCC-TX	4	USBDM
DP	FCC-RX	5	USBDP
VCC33	3V3	6	3V3
MIC	VDD DACVDD	7	DACVDD
MIC	PA0	8	PA0
DACR	DACR	9	DACR
DACL	DACL	10	DACL
SGND	SGND	11	SGND
GND	GND	12	GND
5.0V	5V0V	13	+5.0V

t Side is Digital Ground
y input.

BT401 -- DAC Output

UART is here

Pin 26: Connect to the MCU's RX pin, operates at 3.3V logic level, compatible with 5V.
Pin 25: Connect to a 5V logic level through a 1KΩ resistor in series.

UART---115200bps default

The MUTE pin enables an automatic detection function
When pulled up, high - MUTE, low - operation
When pulled down, low - MUTE, high - operation

23 PB3 MUTE Connect to the enable pin of a power amplifier.

If not used, the pin can be left floating.

Analog Ground and Digital Ground Must Be Shorted at the Power Input: This is very important!!!

1. This IO pin is designed for automatic detection on power-up:
==> If your power amplifier is muted at high level, pull up this IO with a 10K resistor to the power supply [the power supply must be below 5.2V; a 12V supply cannot be directly pulled up].
==> If your power amplifier is muted at low level, pull down this IO with a 10K resistor to ground.
2. The purpose of this pin is to eliminate pop or crackling sounds from some power amplifiers during power-on/off.
3. In Bluetooth mode, this pin enables the power amplifier when playing music and disables it when music stops, reducing background noise. In call mode, it enables the power amplifier.
4. If no power amplifier is used, leave this pin floating. Alternatively, you can control it with your own MCU.

Question 36 36. Complete Functional Introduction to the BT401's ADKEY, with Detailed Description

Answer

The BT401 Bluetooth module supports button control, commonly known as the ADKEY function, located on pin 3, which is PA5.

Note: If the AD key is not used, pin 3 of the module can be left floating, with no need for



pull-up or pull-down.

Button Resistance Value Reference Table				
Resistance Value	Short Press	Long Press	Hold Down	Remarks
0 Ω	Play/Pause	Mode Switch	-	-
3 K Ω	Previous Track	-	Volume Down	-
6.2 K Ω	Next Track	-	Volume Up	-
9.1 K Ω	Mode Switch	-	-	Switches between Music Mode, Bluetooth, and AUX Input
15 K Ω	Volume Down	-	Volume Down	-
24 K Ω	Volume Up	-	Volume Up	-
33 K Ω	Previous Track	-	Fast Backward (Music Mode only)	-
51 K Ω	Next Track	-	Fast Forward (Music Mode only)	-
100 K Ω	Playback Mode Switch	-	-	Toggles between Single Loop and All Loop
220 K Ω	Power On/Off	-	-	-

Question 37	37. Explanation of Bluetooth Audio Local Volume and Synchronization with Phone Volume
Answer	Answer: iOS System: 1. If our chip's current volume is set to level 30, 2. After an iOS device connects to the Bluetooth chip, it will synchronize the volume once. 3. Adjusting the volume buttons on the iOS device will keep the Bluetooth chip's volume in sync with the phone. 4. If the iOS device disconnects, the Bluetooth chip's volume will remain at the previously synchronized level. 5. To reset, power off and power on again. Android System: 1. Android systems are completely different. 2. After an Android device connects to the Bluetooth chip, it will not synchronize volume with the chip. 3. In other words, adjusting the volume on the Android device has no effect on the Bluetooth chip. 4. Essentially, Android devices have two separate volumes: the phone's volume and the Bluetooth chip's DAC volume. General Notes: 1. Only iOS products (e.g., iPhone, iPad) can achieve volume synchronization.

2. Android phones and devices, due to their diversity, cannot synchronize volume. All speakers and earphones on the market behave this way with Android.
3. When firmware needs to handle this, send the command AT+CG00\r\n to disable Bluetooth running in the background.

4.2.12 Setting Whether Bluetooth Runs in the Background [CG]

AT+CG00\r\n	Disable -- Bluetooth does not run in the background.
AT+CG01\r\n	Enable -- Bluetooth runs in the background.

1. For our basic functionality, Bluetooth actually resides in memory for a long time, so even when you're playing a TF card, you can still normally search for Bluetooth.
2. The purpose of this is to ensure Bluetooth data transmission is effective in all modes.
3. To conveniently meet the need of not having "Bluetooth run in the background," we added this command. The setting takes effect after power-off.
4. This allows complete release of Bluetooth resources when playing a TF card, and similarly, Bluetooth data transmission is only effective in Bluetooth mode.

4. After the phone adjusts the volume, the chip actively returns the current synchronized volume, such as the command QA+25.
5. After the phone synchronizes the volume, this volume is memorized and applied to other modes, such as reading a U disk or TF card.

Question 38	38. Detailed Explanation of Bluetooth Data Transparent Transmission and AT Commands—Which Are AT Commands and Which Are Transparently Transmitted?								
Answer	1. Our serial port commands currently support AT commands and also enable full-duplex data transparent transmission between Bluetooth and a phone or other host device. 2. AT commands are active throughout the chip's lifecycle. Once the chip initializes Bluetooth, data transparent transmission runs in the background, supporting AT commands whether connected or not. From the MCU Side: 3. For commands sent by the MCU, anything that is not a standard AT command will be transparently transmitted. Examples: <table border="1"> <thead> <tr> <th>Data sent from the MCU side</th><th>Explanation</th></tr> </thead> <tbody> <tr> <td>AT+CT00\r\n</td><td>This is a normal AT command and will not be transmitted transparently. The Bluetooth chip will process it directly.</td></tr> <tr> <td>AT+CT00</td><td>This is an abnormal command and will be transmitted transparently because it lacks a line break. The Bluetooth chip will also return ER+7.</td></tr> <tr> <td>KT+CT00\r\n</td><td>This will also be transmitted transparently because it does not start with an AT command.</td></tr> </tbody> </table>	Data sent from the MCU side	Explanation	AT+CT00\r\n	This is a normal AT command and will not be transmitted transparently. The Bluetooth chip will process it directly.	AT+CT00	This is an abnormal command and will be transmitted transparently because it lacks a line break. The Bluetooth chip will also return ER+7.	KT+CT00\r\n	This will also be transmitted transparently because it does not start with an AT command.
Data sent from the MCU side	Explanation								
AT+CT00\r\n	This is a normal AT command and will not be transmitted transparently. The Bluetooth chip will process it directly.								
AT+CT00	This is an abnormal command and will be transmitted transparently because it lacks a line break. The Bluetooth chip will also return ER+7.								
KT+CT00\r\n	This will also be transmitted transparently because it does not start with an AT command.								

	1234AT+CM00\r\n	This will also be transmitted transparently because its starting data does not begin with "AT." The AT command is only in the middle, so it will be transmitted transparently.
	12121212121212kkk	This is purely transparent transmission data, so it will be transmitted to the mobile device.
How to handle this transmitted data is up to your ingenuity!		
4. Regarding data transmission from mobile devices, it's easier to understand --- SPP and BLE transparent transmission explanation: All data is transparently transmitted. Even if the phone sends a standard command like AT+CT00\r\n, it is transmitted. The Bluetooth chip does not process it and only outputs it to the MCU via the serial port.		

Question 39	39. Why can't the customer's existing app connect to BLE, and how can it connect or pair with our module chip?
Answer	First, BLE cannot be connected through the phone's system settings; this must be clear. Second, if you have an existing app and want to use our BLE chip, you must check the app's requirements for BLE services and characteristics. They may not match our chip's configuration. 1. If you don't understand BLE, please research online about what BLE is, what services are, and what characteristics are. 2. To check the BLE services and characteristics supported by your original app, follow these steps: (1) Use the "LightBlue" app to test your previous Bluetooth module; it will display the services and characteristics. This tool is provided in our document package. (2) Ask the app's development engineer, who should know. (3) Send us your previous product, and we can help check and modify it.

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